

The automatic deduplication feature of sets is particularly useful when you need to ensure you're working with unique items.

Set Operations: Finding Relationships Between Groups

Sets support powerful mathematical operations that are useful for analyzing relationships between groups of securities:

```
1. tech_stocks = {"AAPL", "MSFT", "GOOGL", "AMZN", "META"}
2. sp500_stocks = {"AAPL", "MSFT", "JNJ", "JPM", "V", "AMZN", "PG"}
3. watchlist = {"TSLA", "AAPL", "NFLX", "DIS"}
4.
5. # Intersection: stocks that are both tech and in S&P 500
6. tech_in_sp500 = tech_stocks.intersection(sp500_stocks)
7. # Result: {"AAPL", "MSFT", "AMZN"}
8.
9. # Union: all stocks from both tech and watchlist
10. combined_universe = tech_stocks.union(watchlist)
11. # Result: {"AAPL", "MSFT", "GOOGL", "AMZN", "META", "TSLA", "NFLX", "DIS"}
12.
13. # Difference: tech stocks not in your watchlist
14. unwatched_tech = tech_stocks.difference(watchlist)
15. # Result: {"MSFT", "GOOGL", "AMZN", "META"}
16.
17. # Symmetric difference: stocks in either tech or watchlist, but not both
18. exclusive_stocks = tech_stocks.symmetric_difference(watchlist)
19. # Result: {"MSFT", "GOOGL", "AMZN", "META", "TSLA", "NFLX", "DIS"}
20.
21.
```

These operations provide efficient ways to analyze relationships between different groups of securities—a common task in portfolio management and strategy development.